

$$U_B = U_{I1} = 4V, U_C = U_{I2} = 1V, i_A = \frac{U_B - U_C}{R}$$

$$U_A = U_B + R \cdot i_A = 7V,$$

$$U_D = U_C - R \cdot i_A = -2V.$$

对 A_3, A_4 ,

$$\left\{ \begin{array}{l} \frac{U_A - U_D}{R} = \frac{U_{B03} = U_{3-} = U_{3+} = 0V.}{R} \end{array} \right.$$

$$\frac{U_{03} - U_4}{R} = \frac{U_4 - U_0}{R}$$

$$U_{4+} = U_D$$

$$\therefore U_0 = 2U_D = -4V.$$

(2) 当开关断开时,

$$\text{对 } A_3, \quad U_{03} = -\frac{1}{RC} \int U_A dt = -\frac{20}{0.05} \int U_A dt.$$

对 A_4 , A_0 实现减法运算.

$$U_0 = \frac{R}{R} 2U_D - U_{03} = -4 + \frac{20}{0.05} \int U_A dt$$

$$= -4 + \frac{140}{0.33} t + U_0(0) = 0.33t$$

$\therefore$ 当 $U_0 = 0$ 时 $t \approx 28.6ms$.